



Role of PC-1 and ACE genes on insulin resistance and cardiac mass in never-treated hypertensive patients. Suggestive evidence for a digenic additive modulation

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Abstract *Background and aim:* Insulin resistance and increased left ventricular mass (LVM) characterize patients with essential hypertension. Some genetic polymorphisms play a role in the modulation of both insulin resistance and LVM. The aim of this work was to investigate whether the PC-1 and ACE genes exert a polygenic control of insulin resistance and LVM in hypertensive patients.

Methods and results: In 158 never-treated hypertensive patients, we evaluated insulin resistance by HOMA index [$\text{insulin } (\mu\text{U/mL}) \times \text{glucose } (\text{mmol/L}) / 22.5$] and LVM by echocardiograms. Genetic polymorphisms were obtained by polymerase chain reaction. PC-1 X121Q genotype carriers (K121Q + Q121Q, $n = 46$) had higher HOMA (3.14 ± 1.28 vs. 2.49 ± 1.25 ; $p = 0.002$) and LVM (137 ± 34 vs. 127 ± 24 g/m²; $p = 0.02$) than K121K patients ($n = 112$). Similarly, ACE DD carriers ($n = 56$) showed higher HOMA (3.94 ± 1.13 vs. 1.98 ± 0.72 ; $p < 0.00001$) and LVM (142 ± 26 vs. 123 ± 25 g/m²; $p = 0.00004$) than XI (ID + II, $n = 102$) patients. When considering both PC-1 and ACE polymorphisms, HOMA ($p < 0.00001$) and LVM ($p = 0.00003$) progressively increased from K121K/XI to X121Q/XI, K121K/DD and X121Q/DD patients. The association of both gene polymorphisms with LVM was no longer significant

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